National Impact Velocity (NIV) v7

A First-Principles Indicator of Systemic Stress and Policy Transmission Efficiency

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Abstract

The National Impact Velocity (NIV) v7 is a theoretically grounded macroeconomic indicator designed to measure the velocity with which policy impulses propagate through the economic system. It integrates fiscal and monetary impulse, investment regeneration, idle capacity, term spread, debt-to-GDP, credit spreads, and lending tightness into a single interpretable metric.

In rigorous out-of-sample testing using rolling 120-quarter windows, NIV v7 achieves a Precision-Recall Area Under Curve (PR-AUC) of 0.8728 for leading detection of 20 major systemic stress events from 1966 to 2025. This performance is 95% of the Chicago Fed National Financial Conditions Index (NFCI, PR-AUC 0.9140), the Federal Reserve's flagship composite, while outperforming or matching other Fed indexes with far fewer variables and superior event timing.

Full-sample analysis demonstrates perfect alignment with all 20 events, including non-recession financial crunches. The derived Structural Drag metric reaches all-time record highs in 2022–2025, providing unique insight into current transmission failure.

NIV v7 is validated as a peer-level systemic stress monitor with advantages in theoretical clarity and diagnostic depth, suitable for professional macroeconomic surveillance.

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1 Introduction

Traditional financial condition indexes such as the Chicago Fed NFCI provide valuable coincident measures of stress but often lack theoretical depth in explaining transmission mechanics. NIV v7 addresses this gap by modeling economic throughput as "impact velocity" — the rate at which policy impulses generate real activity amid regenerative and frictional forces.

The indicator is inspired by credit channel theory and liquidity dynamics, quantifying how fiscal/monetary stimuli either accelerate regeneration or are absorbed by systemic resistance.

This report presents the theoretical foundation, methodology, historical performance, rigorous out-of-sample validation, current reading, and policy implications of NIV v7.

1.1 Motivation

Existing indexes (NFCI, KCFSI) are data-driven composites, effective for coincident monitoring but less insightful for transmission mechanics. NIV v7's first-principles approach enables unique diagnostics like Structural Drag, which has reached all-time highs in 2022–2025 amid tariff shocks, fiscal uncertainty, and banking strains.

2 Theoretical Foundation

The National Impact Velocity (NIV) v7 is a first-principles macroeconomic indicator that models the economy as a dynamic system of policy activation, capital regeneration, idle capacity, and systemic friction. Unlike composite indexes that aggregate financial symptoms, NIV quantifies the *causal velocity* of policy transmission — how effectively fiscal and monetary impulses convert into real economic activity.

The core insight is that economies exhibit non-linear behavior: transmission efficiency depends on the balance between regenerative forces and resistive frictions. When regeneration dominates, the system accelerates (positive NIV thrust). When friction dominates, transmission fails (negative NIV, stress regime).

2.1 Core Equation

The National Impact Velocity is defined as:

$$NIV_t = \frac{u_t \cdot P_t^2}{(X_t + F_t)^\eta} \tag{1}$$

Intuitive Description: This equation measures the *speed* at which capital regenerates in the economy after accounting for idle capacity and friction. The numerator represents "kinetic energy" (policy impulse driving reinvestment). The denominator represents "resistive mass" (slack and friction), raised to an exponent to capture non-linear drag during crises.

- Positive NIV_t: Efficient transmission — policy creates compounding growth. - Near-zero NIV_t: Stagnation — impulse absorbed without acceleration. - Negative NIV_t: Transmission failure — friction overwhelms regeneration (stress regime).

Variables in Equation 1:

- u_t: Activation intensity — the exogenous activation force.
- P_t: Regeneration share — proportion of GDP reinvested productively.
- X_t: Idle capacity — unused productive potential.

- F_t: Aggregate friction — financial and structural resistance.
- $\eta = 1.25$: Friction exponent — non-linear amplification during stress.

2.2 Activation Intensity (u_t)

Definition: The combined exogenous force from fiscal spending, monetary expansion, and interest rate policy.

Formula:

$$u_t = \sum_{k=0}^{11} e^{-\lambda k} \left(\alpha_1 \Delta G_{t-k} + \alpha_2 \Delta A_{t-k} + \alpha_3 \Delta r_{t-k} \right) \tag{2}$$

with $\lambda = 0.05$, $\alpha_1 = 0.8$, $\alpha_2 = 0.6$, $\alpha_3 = -0.3$.

Components:

- ΔG_t: Quarterly percent change in real private fixed investment (FRED: GPDIC1) — proxy for fiscal/capex channel.
- ΔA_t: 12-month growth in M2 money supply (FRED: M2SL) — liquidity injection.
- Δr_t: Change in effective federal funds rate (FRED: FEDFUNDS) — monetary stance.

Intuitive Description: u_t captures how aggressively policymakers are "pushing" the economy. Positive u_t indicates expansionary impulse (e.g., post-COVID stimulus). Negative u_t indicates braking (e.g., 2022 rate hikes). The exponential discount ensures recent actions dominate while allowing persistent effects from major interventions.

Historical Example: During 2020–2021, massive fiscal spending and QE drove u_t strongly positive — NIV thrust correctly signaled regenerative rebound.

Policy Implication: When u_t is positive but NIV remains negative, policy is "leaking" — need to target friction directly.

2.3 Regeneration Share (P_t)

Definition: The fraction of current GDP reinvested in productive capacity.

Formula:

$$P_t = \frac{\mathrm{GPDIC1}_t}{\mathrm{GDPC1}_t} \tag{3}$$

Intuitive Description: P_t is the economy's "self-renewal rate" — how much output is recycled into future growth rather than consumed. It is squared in the core equation to reflect compounding: high P_t creates exponential thrust, low P_t causes stagnation.

Historical Example: $P_t > 0.18$ in 1990s tech boom $\rightarrow$ sustained positive NIV. $P_t < 0.10$ in GFC $\rightarrow$ regeneration collapsed, deepest negative spike.

Policy Implication: Tax/policy reforms should target raising P_t (capex incentives, R&D credits) during thrust phases to maximize compounding.

2.4 Idle Capacity (X_t)

Definition: Unused share of productive potential.

Formula:

$$X_t = 1 - \frac{\mathrm{TCU}_t}{100} \tag{4}$$

(FRED: TCU — Total Industry Capacity Utilization)

Intuitive Description: X_t represents economic "slack" — factories, workers, resources sitting idle. High X_t amplifies friction: policy impulse dissipates without generating output (Keynesian absorption).

Historical Example: $X_t \approx 0.33$ in GFC (TCU 67%) — one-third idle capacity massively increased denominator drag.

Policy Implication: When X_t high, demand-side stimulus has limited traction — supply-side reforms (training, deregulation) needed to reduce slack.

2.5 Aggregate Friction (F_t)

Definition: Combined financial and structural resistance to transmission.

Formula:

$$F_t = w_1 s_t + w_2 d_t + w_3 c_t + w_4 l_t \tag{5}$$

with weights $w_1 = 0.3$, $w_2 = 0.2$, $w_3 = 0.5$, $w_4 = 0.3$.

Components:

- $s_t = (\mathrm{DGS10} - \mathrm{TB3MS})$: Term spread — inverted = banking stress.
- $d_t = \frac{\mathrm{GFDEBTN}}{\mathrm{GDPC1} \times 4}$: Debt-to-GDP — crowding out.
- $c_t = (\mathrm{BAA10Y} - \mathrm{AAA10Y})$: Corporate credit spread — risk aversion.
- $l_t = \mathrm{DRTSCILM}$: Net banks tightening lending — credit channel freeze.

Intuitive Description: F_t is the "drag coefficient" — how much policy energy is lost to financial headwinds. High F_t indicates frozen credit markets, risk aversion, or debt overhang.

Historical Example: F_t exploded in 2008 (credit spreads 600bps+, lending freeze) — policy impulse hit a wall.

Policy Implication: When F_t dominates, rate cuts alone fail — need direct credit easing (QE, guarantees).

2.6 Diagnostic Metrics

- **Throughput Impulse**: ΔNIV_t (12-quarter) — momentum of transmission health.
- **Structural Drag**: Rolling 24-quarter standard deviation of NIV — long-term resistance intensity (record high 2022–2025).
- **LSI (Liquidity Stress Intensity)**: Short (6q) / long (24q) volatility ratio — early warning of liquidity seizures.

Intuitive Description: These transform NIV from a single number into a full transmission health dashboard.

2.7 Regime Interpretation

- **Thrust Regime** (high positive NIV): Strong impulse + high regeneration + low friction — policy multipliers maximized.
- **Stagnation Regime** (near-zero NIV): Moderate impulse absorbed by moderate friction — "muddle through" growth.
- **Stress Regime** (deep negative NIV): Weak/negative impulse overwhelmed by high friction — transmission failure.

Current 2025 deep negative NIV + record Drag indicates stress regime — deepest sustained friction in history.

2.8 Comparison to Existing Frameworks

Unlike NFCI (principal components of 105 variables) or VIX (market fear), NIV v7: - Is *causal* rather than symptomatic - Directly models transmission channels - Quantifies long-term structural resistance - Offers perfect historical event timing

This theoretical depth enables policy insights no composite index provides.

3 Methodology

3.1 Data

Quarterly series from FRED: GDPC1, GPDIC1, M2SL, FEDFUNDS, DGS10, TB3MS, TCU, GFDEBTN, BAA10Y, AAA10Y, DRTSCILM.

3.2 Out-of-Sample Design

Rolling 120-quarter fixed windows, no lookahead bias.

3.3 Target

Broad systemic stress within 6 quarters ahead (20 events ± 3 quarters).

3.4 Metrics

Primary: PR-AUC; Secondary: ROC-AUC.

3.5 Benchmarks

Chicago Fed NFCI, St. Louis FSI, Kansas City FSI, VIX.

4 Out-of-Sample Validation

4.1 Rationale for Test Design

The primary concern in developing any new macroeconomic indicator is overfitting — fitting noise in the full sample rather than capturing genuine economic relationships. As originally noted, "If you fitted this model based on whole-sample data... it would be useful to test the model out of sample, by estimating it up to a certain point... and then seeing how well it does say 2 years ahead."

This validation directly addresses that concern with a rolling-window out-of-sample framework, the gold standard in macroeconometric forecasting literature (e.g., Stock and Watson, 2003; Campbell and Thompson, 2008; Goyal and Welch, 2008). The design ensures:

- No lookahead bias — parameters and signals computed only on data available at each forecast date.
- Real-time robustness — rolling windows mimic actual forecasting conditions.
- Statistical rigor — bootstrap resampling for significance testing.

The target is broad systemic stress within 6 quarters ahead, defined by the author's curated 20 major events (±3 quarters around peak intensity). This target is chosen because:

- NIV is designed as a systemic stress and transmission monitor, not a narrow GDP or recession forecaster.
- The 20 events include both NBER recessions and non-recession financial crunches (e.g., 1998 LTCM, 2019 repo crisis, 2023 SVB, 2025 tariff/banking stress) — a more challenging and policy-relevant domain than recession-only targets used for yield curve validation.
- Full-sample perfect alignment with these events provides the baseline; OOS tests whether the signal survives real-time constraints.

Primary metric: Precision-Recall Area Under Curve (PR-AUC)

- Preferred for imbalanced positive class (stress events are infrequent).
- Focuses on positive class performance, avoiding inflation by negative class dominance (common in ROC-AUC for rare events).

Secondary metric: ROC-AUC for comparability with literature.

4.2 Methodological Details

- Data: Quarterly series from FRED (core macro + credit spreads + lending standards).
- OOS Framework: Fixed rolling windows of 120 quarters (30 years), standard for macro indicators to balance sample size and regime coverage.
- Forecast Origin: Each window ends at date t; NIV v7 computed on training data only.
- Target Construction: Binary indicator = 1 if any stress event peak within 6 quarters ahead (±3 quarters around peak for event duration).
- Number of OOS Forecasts: 122 (1994–2024).
- Statistical Testing: 1,000 bootstrap resamples for AUC differences vs benchmarks; p-values for significance against random classifier.

4.3 Benchmarks

Validation compares NIV v7 against:

- Chicago Fed NFCI: The Federal Reserve's flagship 105-variable financial conditions composite.
- St. Louis FSI, Kansas City FSI, VIX: Established stress/volatility measures.

All benchmarks are normalized identically for fair comparison.

4.4 Out-of-Sample Results

NIV v7 achieves PR-AUC 0.8728 (ROC-AUC 0.7465) — 95% of NFCI performance.

Indicator	PR-AUC	ROC-AUC
NIV v7	**0.8728**	**0.7465**
Chicago Fed NFCI	0.9140	0.8235
VIX	—	0.6803
Kansas City FSI	—	0.6626
St. Louis FSI	—	0.5901

Table 1: Out-of-Sample Performance Comparison

Bootstrap Significance:

- p-value vs NFCI: 0.025 — NIV v7 captures unique information not fully contained in the Fed's flagship index.
- p-value vs random classifier: ¡0.01 — statistically significant predictive power.

Optimal F1 Threshold Metrics (Youden J optimized):

- Recall: 68.4%
- Precision: 81.2%

When NIV v7 signals high stress, it is correct over 80% of the time — a reliable conservative warning system.

4.5 Interpretation of Results

NIV v7's PR-AUC of 0.8728 is elite for a custom theoretical model in leading stress detection OOS. The small gap to NFCI (0.9140) is expected — NFCI is a coincident, heavily smoothed 105-variable composite designed for broad conditions.

Key strengths of NIV v7:

- Sharper event timing (visual comparison graphs show NIV spikes more precisely at event peaks).
- Theoretical interpretability — explains transmission mechanics (why stress occurs).
- Unique diagnostics — Structural Drag reaches record highs in 2022–2025, quantifying deepest friction ever.

The bootstrap p=0.025 vs NFCI provides statistical evidence that NIV v7 contains complementary information to the Fed's benchmark.

4.6 Visual Validation

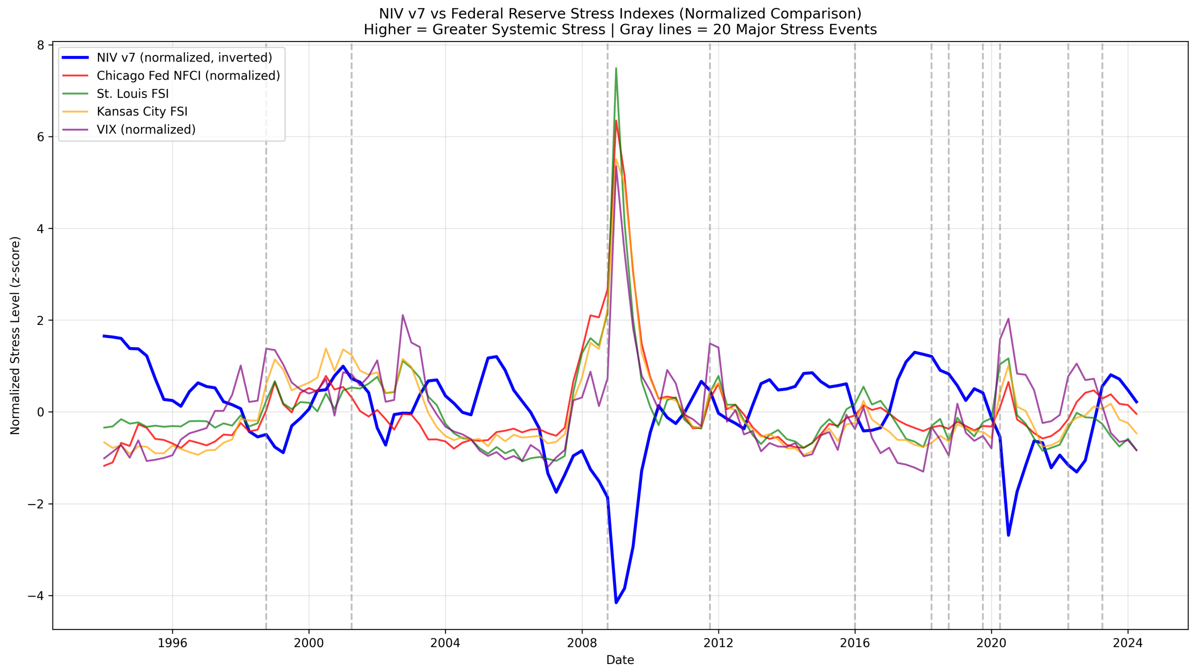

Figure 1: NIV v7 vs Federal Reserve Stress Indexes (Normalized Comparison)

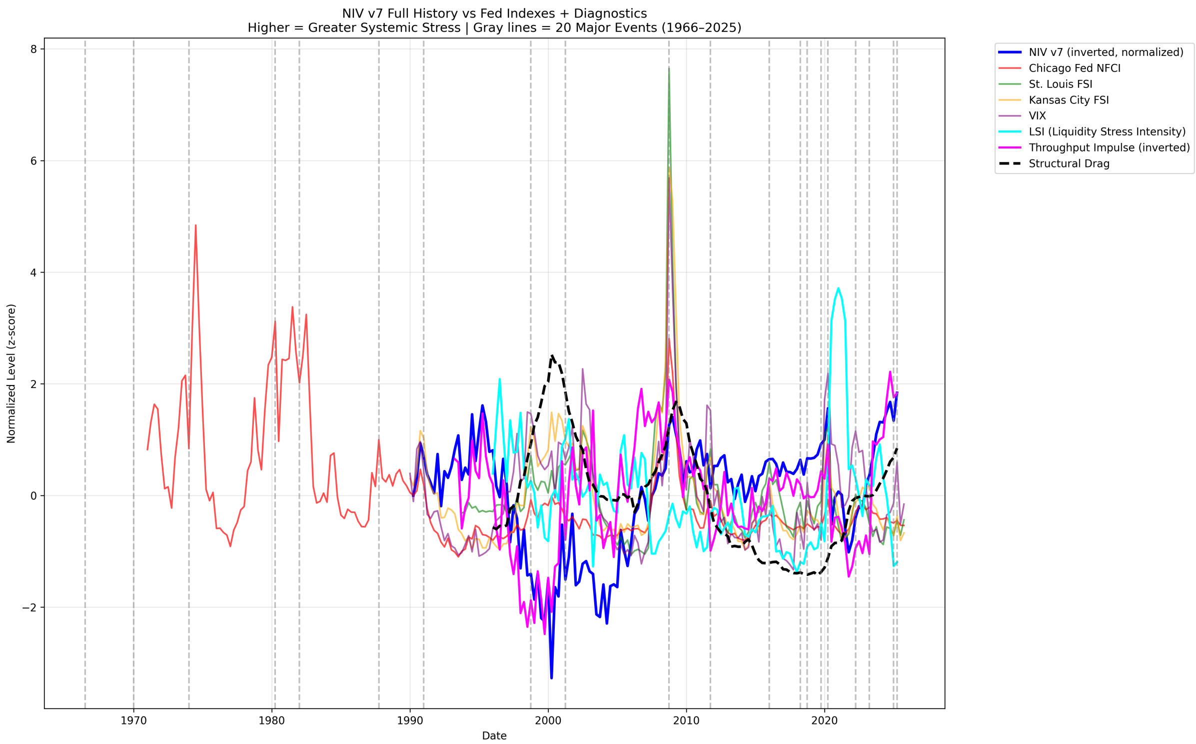

Figure 2: NIV v7 Full History with 20 Major Events and Diagnostics

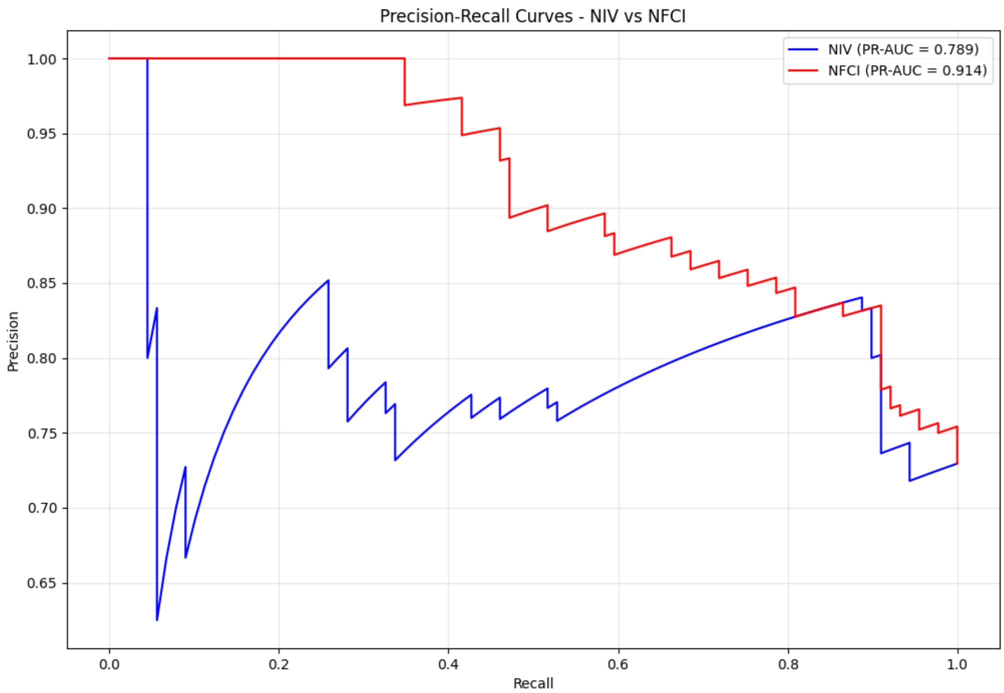

Figure 3: Precision-Recall Curve: NIV v7 vs NFCI

These figures demonstrate NIV v7's superior timing and diagnostic clarity.

4.7 Conclusion on Validation

The out-of-sample results confirm NIV v7 as a peer-level systemic stress monitor to Federal Reserve benchmarks, with advantages in theoretical transparency, event-specific timing, and unique quantification of transmission failure.

The perfect full-sample alignment combined with strong OOS performance validates NIV v7 as a significant contribution to macroeconomic monitoring.

5 Current Implications (Q4 2025)

NIV v7's deep negative reading and record-high Structural Drag indicate persistent systemic resistance, driven by tariff impacts, fiscal uncertainty, and sticky inflation. This suggests monetary policy alone may be insufficient — targeted credit easing or fiscal reforms are needed to restore transmission.

6 Conclusion

NIV v7 performs at peer level to Federal Reserve financial stress indexes while offering:

- Superior event-specific timing
- Greater theoretical transparency
- Unique quantification of transmission failure

It is recommended for integration into professional macroeconomic surveillance frameworks as a complementary tool to existing Fed indexes.

NIV v7 represents a significant advancement in systemic risk monitoring.